Joseph Wardle - Pipeline Technical Director

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Work Experience

Pipeline Technical Director - Student Short Films October 2025 - Present
Brigham Young University Center for Animation *Provo, UT*

- Develop artist tools and maintain a **USD production pipeline** supporting **50+ artists** across modeling, previs, animation, FX/CFX, shading, and rendering on Linux and Windows.
- Built a **Python/Qt playblast viewer used across every department**, with scrubbing, looping, and multi-clip playback; reduced dailies preparation through department-specific burn-ins and submission to editorial and **ShotGrid**.
- Developed a lightweight **ShotGrid API wrapper** to query, create, and update shots, assets, and playlists, supporting artist tools that reduced production-data authoring errors.
- Developed **Maya previs tools** to batch-playblast shots with handles, review them consecutively, and export individual clips or a combined movie through the shared submission workflow.
- Built an **FX/CFX wedge comparison interface** for farm-generated flipbooks and renders, with parameter burn-ins, grid comparison, individual inspection, and one-click selection of simulation caches.
- Automated **Houdini lookdev turntables** with a cyclorama, color checker, and reference spheres, farm rendering, **color-managed movie generation**, and ShotGrid review submission.
- Implemented **immutable USD publishes** and pinned layer versions at **Tractor** submission, keeping render dependencies consistent when assets were republished mid-render.

Web Developer March 2025 - November 2025
Brigham Young University - Computer Science Department *Provo, UT*

- Refactored **Django/PostgreSQL data models** for TA/RA hiring to represent coursework, faculty assignments, and employment history, helping faculty track appointments and information used to determine pay.
- Developed **REST APIs and administrative interfaces** using **Django ORM** for database access and **Celery/Redis** for background tasks and scheduling.
- Built **GitHub Actions** workflows for testing, container builds, and deployment; used **Docker Compose** for local development and **Jira** for project tracking.

Selected Projects

Cenote - GPU Path Tracer and Hydra Render Delegate github.com/joseph-wardle/cenote

- Built a **Rust/Vulkan** GPU path tracer and **Hydra render delegate** integrating with usdview, husk, and Houdini; implemented **OpenPBR** shading in Slang.

Film Grain Synthesis (github.com/joseph-wardle/film_grain) August 2025

- Reimplemented a published film-grain algorithm in **Rust** with **Rayon** and **wgpu** compute; benchmarked the CPU implementation across approximately 7,200 configurations, measuring a **median 72× speedup** over the reference C implementation.

Education

Brigham Young University Provo, UT
Bachelor of Science, Computer Science (Animation & Games Emphasis) *August 2023 - Expected May 2027*

- **GPA: 3.95.** Relevant coursework: Data Structures, Multithreading, Linear Algebra, Modeling, Rigging, Shading, FX.

Skills

- **Programming:** Python, C++23, Rust, SQL, MEL, Bash
- **Data & Services:** PostgreSQL, Django ORM, REST APIs, Celery, Redis
- **Media & Pipeline:** Qt/PySide2, FFmpeg, ShotGrid, OCIO, USD, Pixar Tractor
- **DCCs:** Maya, Houdini, Nuke, Substance Painter, Blender
- **Development:** Linux, Windows, Git, Perforce, GitHub Actions, Docker, Jira